Basic Information

Experiment No.: 13

Date: 11-10-2026

Title: Letter Combinations of a Phone Number

Objective / Aim

To write a Java program that generates all possible letter combinations represented by the digits from "2" to "9" using the telephone keypad mapping.

System Requirements

Hardware

- Computer/Laptop
- Minimum 4 GB RAM

Software

- Java JDK
- IntelliJ IDEA or any Java IDE
- Operating System: Windows/Linux

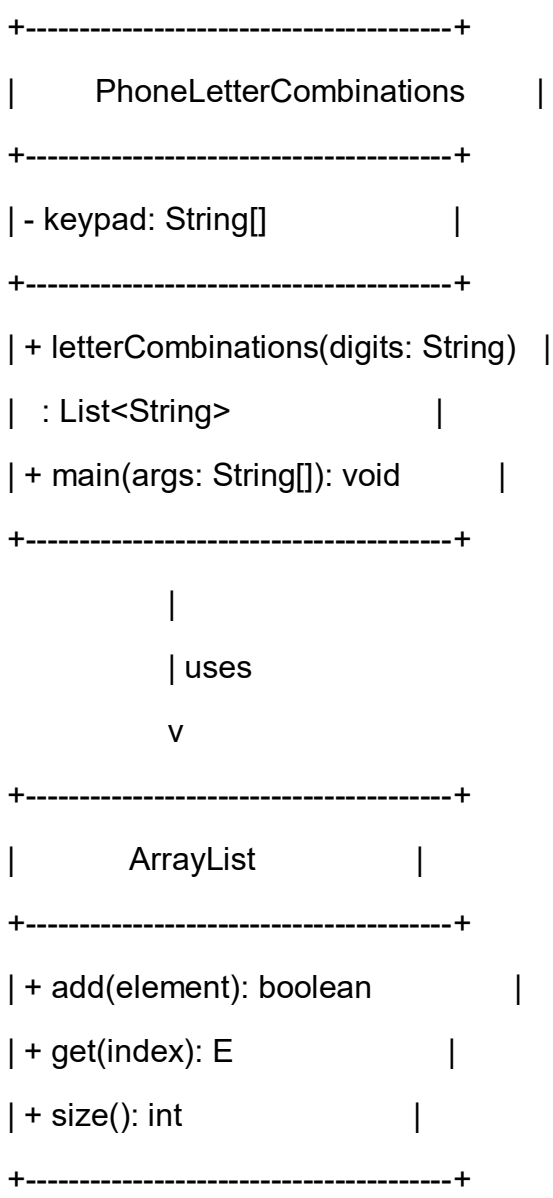
Algorithm / Logic Description

1. Read a string containing digits from "2" to "9".
2. Create a mapping between digits and their corresponding letters.
3. If the input is empty, return an empty list.
4. Start with an empty combination.
5. For each digit, combine every existing combination with its corresponding letters.

6. Repeat until all digits are processed.

7. Display all possible letter combinations.

📐 UML Class Diagram



💻 Source Code

```
import java.util.ArrayList;
import java.util.List;
import java.util.Scanner;

public class PhoneLetterCombinations {

    // Mapping of digits to telephone keypad letters
    static String[] keypad = {
        "", "", "abc", "def", "ghi",
        "jkl", "mno", "pqrs", "tuv", "wxyz"
    };

    // Generate all possible letter combinations
    public static List<String> letterCombinations(String digits) {

        List<String> combinations = new ArrayList<>();

        // Return an empty list if input is empty
        if (digits.isEmpty()) {
            return combinations;
        }

        // Begin with an empty combination
        combinations.add("");

        // Process each digit
        for (int i = 0; i < digits.length(); i++) {
```

```
            int digit = digits.charAt(i) - '0';
            String letters = keypad[digit];

            List<String> newCombinations = new ArrayList<>();

            // Combine existing strings with each mapped letter
            for (String combination : combinations) {
                for (char letter : letters.toCharArray()) {
                    newCombinations.add(combination + letter);
                }
            }

            combinations = newCombinations;
        }

        return combinations;
    }

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        // Read the digit string
        System.out.print("Enter digits (2-9): ");
        String digits = sc.nextLine();

        // Display all combinations
        List<String> result = letterCombinations(digits);
```

```
        System.out.println("Letter combinations: " + result);

        sc.close();
    }
}
```

🔨 Compilation & Execution Commands

Save the file as:

PhoneLetterCombinations.java

Compile:

javac PhoneLetterCombinations.java

Run:

java PhoneLetterCombinations

📄 Output Screen

Sample Input

Enter digits (2-9): 23

Sample Output

Letter combinations: [ad, ae, af, bd, be, bf, cd, ce, cf]

📌 Result

The Java program was successfully implemented to generate all possible letter combinations for the given phone number digits using the Java Collection Framework and "ArrayList".